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Statement on how to interpret the QPS qualification on 'Acquired antimicrobial resistance genes'

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Abstract

The qualified presumption of safety (QPS) approach was developed to provide a regularly updated generic pre-evaluation of the safety of microorganisms intended for use in the food or feed chains. Safety concerns identified for a taxonomic unit (TU) are, where possible, confirmed at the species/strain or product level and reflected by 'qualifications' which should be assessed at strain and/or product level by EFSA's Scientific Panels. The generic qualification 'the strains should not harbour any acquired resistance (AMR) genes to therapeutic antimicrobials' applies to all QPS bacterial TUs. The EFSA 'guidance on the characterisation of microorganisms in support of the risk assessment of products used in the food chain' has aligned the requirements on the approach to assess the qualification related to AMR genes. In this statement, the terms 'intrinsic' and 'acquired' AMR genes were defined for the purpose of EFSA's risk assessments, and they apply to bacteria used in the food and feed chains. A bioinformatic approach is proposed for demonstrating the 'intrinsic'/'acquired' nature of an AMR gene. This approach was applied to different *Bacillus* sp. and the bioinformatic pipeline for the analysis was made publicly available. All AMR genes that confer resistance to therapeutic antimicrobials, found as hits, need to be considered as hazards (for humans, animals and environment) and need further assessment. Genes identified as responsible for 'intrinsic' resistance could be considered as being of no concern in the frame of the EFSA risk assessment. 'Acquired' AMR genes resulting in a resistant phenotype should be considered as a concern. If the presence of the 'acquired' AMR gene is not leading to phenotypic resistance, further case-by-case assessment is necessary.

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Addendum: Data requirements for applicants were clarified (e.g. clonality issue, information on the distribution of a gene). This addendum does not materially affect the contents or outcome of this scientific output. To avoid confusion, the original version of the output has been removed from the *EFSA Journal*, but is available on request.

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Summary

To support the risk assessment of microorganisms carried out by the different EFSA risk assessment areas, EFSA has introduced the qualified presumption of safety (QPS) concept for microorganisms as a generic safety pre-assessment approach that covers safety concerns for humans, animals, and the environment. In the QPS concept, a safety assessment of a defined taxonomic unit (TU) is performed independently of the legal framework under which the application is made during a market authorisation process.

Safety concerns for a TU are, where possible, reflected as 'qualifications', which should be assessed at strain and/or product level by the respective EFSA's Scientific Panel. The generic qualification 'the strains should not harbour any acquired resistance genes to therapeutic antimicrobials' applies to all QPS bacterial TUs.

The available definitions of 'acquired' and 'intrinsic' antimicrobial resistance (AMR) genes and, eventually, their distinction have resulted in certain ambiguity in the interpretation of that generic QPS qualification, both by the applicants submitting the dossier of a regulated product and the EFSA Units responsible for the respective risk assessment. The QPS working group and BIOHAZ Panel have decided to clarify the concepts of 'acquired' and 'intrinsic' AMR genes, considering the current body of knowledge on the genetic basis of these types of resistance and their clinical relevance. This assessment has been based on expert knowledge and published data.

QPS bacterial TUs are used in different areas of EFSA risk assessment: feed additives, pesticides (plant protection products – PPPs), food ingredients and packaging, novel foods (NFs) and genetically modified organisms (GMOs). The "EFSA guidance on the characterisation and risk assessment of microorganisms used in the food chain (EFSA Scientific Committee, 2025) has aligned the requirements on the characterisation of microorganisms for applications underpinning the assessment of microorganisms in the different areas of the food chain". The investigation of the presence of genes involved in the resistance to therapeutic antimicrobials is included in this guidance and their assessment is based on the definition of 'intrinsic' and 'acquired' antimicrobial resistance gene as defined in this Statement.

In this statement, the terms 'intrinsic' and 'acquired' AMR genes were defined for the purpose of EFSA's risk assessments, and they apply to bacteria used in the food and feed chains. A bioinformatic approach is proposed for demonstrating the 'intrinsic'/'acquired' nature of an AMR gene. As an example aiming to implement this approach, a bioinformatics pipeline was developed and applied to *Bacillus* species that are most frequently subject of applications for regulated products submitted to EFSA (2024a). The pipeline for the analysis of gene distribution in microbial species is available at the EFSA Knowledge Junction at the link (https://zenodo.org/records/12608405).

All AMR genes encoding resistance to therapeutic antimicrobials are, in principle considered as hazards and need further assessment. Genes conferring 'intrinsic' resistance could be considered as of no concern in the frame of the EFSA risk assessment purpose. 'Acquired' AMR genes coinciding with the corresponding antimicrobial phenotypic resistance should be considered as a hazard/concern. If the presence of the 'acquired' AMR gene is not leading to phenotypic resistance, further case-by-case assessment is necessary.

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1. Introduction

A wide variety of microorganisms are intentionally used at different stages of the food and feed chains. In the context of applications for pre-market authorisation, the European Food Safety Authority (EFSA) is requested to assess the safety of microorganisms used 'as such' or as production organisms for food and feed additives, food enzymes, and food flavourings, plant protection products (PPPs), novel foods (NFs) or genetically modified microorganisms (GMMs). In each area, the risk assessment falls under specific legislations.

To support the risk assessment of microorganisms carried out by the different EFSA risk assessment areas, in 2004 EFSA introduced the qualified presumption of safety (QPS) concept for microorganisms as a generic safety pre-assessment approach that covers safety concerns for humans, animals, and the environment (EFSA, 2005). In the QPS concept, a safety assessment of a defined taxonomic unit (TU) is performed independently of the legal framework under which the application is made during a pre-market authorisation process. Strains belonging to QPS TUs still require an assessment based on data analysis by the relevant EFSA Scientific Panel, but the QPS status facilitates this evaluation (EFSA BIOHAZ Panel, 2023).

Safety concerns for a TU are, where possible, reflected as 'qualifications', which should be assessed at strain and/or product level by EFSA's Scientific Panels. The generic qualification 'the strains should not harbour any acquired resistance genes to therapeutic antimicrobials' applies to all QPS bacterial TUs (EFSA BIOHAZ Panel, 2023).

The "EFSA guidance on the characterisation and risk assessment of microorganisms used in the food chain (EFSA Scientific Committee, 2025) has aligned the requirements on the characterisation of microorganisms for applications underpinning the assessment of microorganisms in the different areas of the food chain". The investigation of the presence of genes involved in the resistance to therapeutic antimicrobials is included in this guidance and their assessment is based on the definition of 'intrinsic' and 'acquired' antimicrobial resistance gene as defined in this Statement.

The term 'therapeutic antimicrobials' used within this document but also within the "EFSA guidance on the characterisation and risk assessment of microorganisms used in the food chain" (EFSA Scientific Committee, 2025), refers to antimicrobials of medical and veterinary importance for humans and animals as defined by the World Health Organization (WHO) and World Organisation for Animal Health (WOAH), respectively ('medically important antimicrobials' (WHO, 2024[1] and future updates) and 'veterinary critically important antimicrobial agents, veterinary highly important antimicrobial agents and veterinary important antimicrobial agents' (WOAH, 2024[2] and future updates).[3]

1.1. Background and Terms of Reference as provided by the requestor

The overarching mandate related to the QPS activities is included in the latest QPS opinion (EFSA BIOHAZ Panel, 2023). The scope of this Panel statement is very specific and related to the clarification of a QPS qualification that may be considered linked to part of ToR2 (…"*to verify if the qualifications still effectively exclude safety concerns*.").

1.2. Scope of this document

The generic qualification for bacterial TUs that 'the strains should not harbour any acquired antimicrobial resistance genes to therapeutic antimicrobials' applies to all QPS bacterial TUs (EFSA BIOHAZ Panel, 2023). This document focuses on QPS bacteria, but the principles described may also apply to non-QPS bacteria as per "EFSA guidance on the characterisation and risk assessment of microorganisms used in the food chain" (EFSA Scientific Committee, 2025). The use of a QPS microorganism, viable/inactivated, or its product(s) should not add to the overall pool of AMR genes nor increase the spread of AMR. The potential of an AMR gene to be transferable/mobilisable is a key element in this respect. All bacterial genomes contain a number of mobile or mobilisable genetic elements that, with different frequencies, may be involved in horizontal gene transfer (Andam and

[1] WHO (World Health Organization). 2024. WHO's List of Medically Important Antimicrobials: A risk management tool for mitigating antimicrobial resistance due to non-human use. Available online: https://cdn.who.int/media/docs/default-source/gcp/who-mia-list-2024-lv.pdf?sfvrsn=3320dd3d_2.

[2] WOAH (World Organisation for Animal Health), WOAH List of Antimicrobial Agents of Veterinary Importance (June 2024). Available online: https://www.woah.org/app/uploads/2021/06/amended-91gs-tech-03-amr-working-group-report-en.pdf.

[3] Classification of antimicrobials is reliant on the provision of guidance by relevant bodies (World Health Organization (WHO), World Organisation for Animal Health (WOAH), European Medicines Agency (EMA), European Commission (EC), etc.).

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Gogarten, 2011). However, the analysis of mobile or mobilisable genetic elements in the whole genome sequence (WGS) of a bacterium and/or experiments for testing transferability of genes are not considered applicable as a basis for the EFSA safety assessment of AMR genes, as any genetic element can be transferred at a higher or lower frequency. It has been recognised that low frequency events are difficult to reproduce in lab-designed experiments. Although improvements have been obtained by recent developments in whole genome sequencing technology, currently analysing the WGS data for the presence of mobile or mobilisable genetic elements may still lead to false positive and/or false negative results.

The definitions and principles included in this document in relation to AMR genes are intended to contribute towards the assessment of bacteria notified to EFSA for pre-market authorisation to be used in the food and feed chains.

This document aims to clarify the definitions of 'intrinsic' and 'acquired' AMR genes and provide some general principles on the risk assessment of AMR genes present in QPS bacterial TUs. The definitions and principles are intended to harmonise and support the risk assessment performed in the different EFSA risk assessment areas. They should therefore be generally applicable and of practical use.

The terms 'intrinsic' and 'acquired' AMR genes are in this document specifically defined for the purpose of EFSA's risk assessments and they apply to bacteria used in the food and feed chains.

The tools and methodologies recommended in this document are formulated based on the current knowledge in an area that is constantly evolving.

In this document the following topics are covered:

1) Definition of the terms 'intrinsic' and 'acquired' AMR genes under the scope of EFSA's risk assessment.
2) Definition of general principles and criteria for guiding the EFSA risk assessment of AMR genes.

2. Data and Methodologies

As this updated Panel statement is intended to further clarify certain concepts and to support the level of applicability of a QPS qualification associated with all bacteria included in the QPS list, expert knowledge and data from the literature were used to support the content of this BIOHAZ Panel Statement. For literature data, relevant databases, such as PubMed, Web of Science, CAB Abstracts or Food Science Technology Abstracts (FSTA) and Scopus, were used, based on expert judgement.

In line with EFSA's policy on openness and transparency, and to receive comments on its work from the scientific community and stakeholders, EFSA engages in public consultations on key issues for which external input would be desirable.

EFSA's Panel on Biological Hazards (BIOHAZ) carried out in 2023 an open consultation to receive input from interest parties on this Panel statement on how to interpret the QPS qualification for all QPS bacterial taxonomic units '*the strains should not harbour any acquired resistance genes to therapeutic antimicrobials*'. This public consultation gave stakeholders the opportunity to provide any relevant evidence not included in the scientific assessment, and feedback on the methods, results, conclusions, and recommendations. All details may be found in Annex A – Outcome of the Public consultation on the draft Statement on how to interpret the QPS qualification on 'Acquired antimicrobial resistance genes'.

3. Assessment

3.1. Definition of 'intrinsic' and 'acquired' antimicrobial resistance genes under the scope of EFSA's risk assessment

3.1.1. Definition of 'intrinsic' and 'acquired' AMR genes

This statement focuses on the hits corresponding to AMR genes detected when interrogating the WGS data of the bacterial strain under assessment for the presence of AMR genes (as explained in the EFSA WGS Statement; EFSA, 2024b). These AMR genes need to be classified as 'intrinsic' or 'acquired' for the purpose of EFSA's risk assessments, and they apply to bacteria used in the food and feed chains.

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- 'Intrinsic' AMR genes are those inherent to strains of a bacterial species, limiting the action of antimicrobial agents and thereby allowing them to survive and multiply in presence of the antimicrobial agents. An AMR gene is considered 'intrinsic' when it is shared by the vast majority of wild-type strains of the same species[4] and is restricted to the chromosome (see Section 3.1.2).
- The term 'acquired' AMR gene is interpreted as a resistance gene novel for the strain under assessment, acquired through horizontal transfer, enabling a bacterial strain to survive or multiply in presence of concentrations of an antimicrobial agent higher than those that inhibit the growth of the majority of wild-type strains of the same species without this AMR gene (see Section 3.2). 'Acquired' AMR genes could be integrated in the bacterial chromosome or harboured on a separate genetic element.

3.1.2. Genome based identification of 'intrinsic' AMR genes

The risk assessment of AMR genes should contain a bioinformatic analysis of the WGS of the bacterial strain under investigation. For this, the requirements for sequencing and data quality control included in the most recent update of the "EFSA Statement on WGS analysis of microorganisms intentionally used in the food chain" are recommended to be followed (EFSA, 2024b and future updates). The document sets the principles and minimum criteria for the conducting and reporting of the AMR analysis at the time of drafting the document and provides guidance on the specifications for the bioinformatic analysis.

EFSA requires the use of two regularly updated databases to investigate the possible presence of AMR genes in genomes (EFSA, 2024b and future updates). The ENA or NCBI protein ID or nucleotide accession numbers should be provided for each of the AMR hits.

To demonstrate the 'intrinsic' nature of an AMR gene, a sufficient number of high-quality complete genomes of strains of the single species should be interrogated for the presence of the AMR gene under investigation (see next paragraph for more details). The genome sequences present in publicly available databases (ENA, SRA from NCBI, DDBJ)[5] or provided by the applicant could be used.

Generally, the analysis should start with all genome sequences of the strains of the species available in the databases/repositories (at least ENA or SRA should be included). The databases used and the year of release should be documented. The reason why any strains would not be included for further analysis needs to be documented. If fewer than 30 strains are available, it is recommended to include sequences of more strains based on the applicant's own research. The quality of the WGS and the assembly should be provided according to EFSA (2024b and future updates). RefSeq sequences[6] as available in the NCBI database[7] are curated and would not need further quality control. The species identity of the strains should be verified based on the available WGSs according to the specifications set in the WGS Statement (EFSA, 2024b and future updates). This includes an identification preferentially based on digital DNA–DNA hybridization (dDDH) and/or average nucleotide identity (ANI) or phylogenomic methods. The isolates of the same species should be epidemiologically and/or ecologically unrelated. The non-clonal relationship can, in most cases, be demonstrated by providing a phylogenomic tree based on SNP analysis, ANI distance, pangenome analysis of the available WGSs. This phylogenomic information should be carefully interpreted based on species specific taxonomic information and complemented, where possible, with metadata of the strains included in the analysis (e.g. source and isolation) to avoid any bias in the analysis.

The assessment should be based on an identity search (e.g. BLAST) of the genes homologous to the AMR gene under investigation in the genome sequences selected as described above. A positive control gene expected to be present in all strains of the target group (e.g. a DNA replicon initiator) must be included. In general, query sequence hits with at least 80% identity (at the protein level or nucleotide level as reported in the database) and 70% length of the subject sequence should be taken

[4] For species with Standing in Nomenclature - LPSN (List of Prokaryotic Names with Standing in Nomenclature). Available online: https://lpsn.dsmz.de.

[5] EMLB-EBI (which hosts ENA), NCBI (which hosts SRA) and DDBJ work closely together, and are interconnected, as partners in the International Nucleotide Sequence Database Collaboration (INSDC) https://www.insdc.org/.

[6] NCBI Reference Sequence Database. The Reference Sequence (RefSeq) collection provides a comprehensive, integrated, non-redundant, well-annotated set of sequences, including genomic DNA, transcripts, and proteins. https://www.ncbi.nlm.nih.gov/refseq/about/.

[7] Pruitt K, Murphy T, Brown G, et al. RefSeq Frequently Asked Questions (FAQ) 2010 Nov 15 [Updated 2020 Jan 31]. In: RefSeq Help [Internet]. Bethesda (MD): National Center for Biotechnology Information (US); 2011. Available online: https://www.ncbi.nlm.nih.gov/books/NBK50679/.

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into account (EFSA, 2024b and future updates). Based on a proper analysis performed as described above, an AMR gene can be considered 'intrinsic' if it is present in the chromosome of the vast majority of the wild-type strains of a bacterial species. The interpretation and conclusion will be species and gene-dependent and will be made on a case-by-case basis, performed in the different areas of EFSA risk assessment.

Aiming to implement the EFSA approach to discriminate between 'intrinsic' and 'acquired' AMR genes, a bioinformatics pipeline was developed and applied to the species of the genus *Bacillus* that are most frequently subject of applications for regulated products submitted to EFSA. The results are presented as a catalogue of genes indicating their abundance and distribution among complete and confirmed genomes publicly available for each species (EFSA, 2024a). The pipeline for the analysis of gene distribution in microbial species is available at the EFSA Knowledge Junction at the link (https://zenodo.org/records/12608405).

3.2. General principles and criteria for guiding the EFSA risk assessment of AMR genes

All identified hits corresponding to genes associated with resistance to therapeutic antimicrobials (see Section 3.1.2) are, in principle, considered as hazards and need further assessment. This assessment is based on the following general principles:

- The discrimination between 'intrinsic' or 'acquired' AMR genes (see Section 3.1.2). Hits identifying 'intrinsic' resistance will be considered as of no concern in the frame of the EFSA risk assessment purpose.
- Any 'acquired' AMR gene with corresponding antimicrobial phenotypic resistance should be considered as a hazard/concern. This phenotypic resistance enables a bacterium to survive or multiply in the presence of an antimicrobial agent at concentrations higher than those tolerated by other strains of the same species without this acquired resistance. This analysis is based on phenotypic antimicrobial susceptibility testing, with MIC determination (the MIC cut-off values (EFSA Scientific Committee, 2025)).
- If the presence of the 'acquired' AMR gene does not lead to phenotypic resistance, further assessment is considered necessary. A case-by-case assessment is necessary to evaluate the likelihood that the AMR gene will be expressed.

Phenotypic antimicrobial resistance not directly linked to the presence of a known AMR mechanism should on a case-by-case be assessed in the context of the EFSA risk assessment.

4. Conclusions

- The terms 'intrinsic' and 'acquired' AMR genes were defined for the purpose of EFSA's risk assessments, and they apply to bacteria used in the food and feed chains. A bioinformatic approach is proposed for demonstrating the 'intrinsic'/'acquired' nature of an AMR gene.
- All AMR genes conferring resistance to therapeutic antimicrobials, found as hits in the bioinformatic analysis, should be considered as hazards for humans, animals and environment, and need further assessment. Hits identifying genes conferring 'intrinsic' resistance could be considered as 'of no concern' in the frame of the EFSA risk assessment. 'Acquired' AMR genes with corresponding antimicrobial phenotypic resistance should be considered as a concern. If the presence of the 'acquired' AMR gene is not leading to phenotypic resistance, further case-by-case assessment is necessary.

5. Recommendation

- It is recommended that this document is regularly reviewed and revised as required in response to any developments in the knowledge about, and the tools for defining and discriminating, 'intrinsic' and 'acquired' AMR genes.

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Abbreviations

AMR	antimicrobial resistance
ANI	average nucleotide identity
BIOHAZ Panel	EFSA Panel on Biological Hazards
BLAST	Basic Local Alignment Search Tool
CAB	Centre for Agriculture and Biosciences
dDDH	digital DNA–DNA hybridization
DDBJ	DNA Data Bank of Japan
ENA	European Nucleotide Archive
FSTA	Food Science Technology Abstracts
GMMs	genetically modified microorganisms
GMOs	genetically modified organisms
ID	identification
MIC	Minimal Inhibitory Concentration
NCBI	National Center for Biotechnology Information
NFs	novel foods
PPPs	plant protection products
QPS	qualified presumption of safety
SRA	Sequence Read Archive
ToR	Terms of Reference
TU(s)	taxonomic Unit(s)
WGS	whole genome sequence
WHO	World Health Organization

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Annex A – Outcome of the Public consultation on the draft Statement on how to interpret the QPS qualification on 'Acquired antimicrobial resistance genes'

Annex A is available under the Supporting Information section on the online version of the scientific output.

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